

Predicting ED-to-Ward Waiting Times in Singapore Public Hospitals

Group 6_P2

Colab Notebook:

<https://colab.research.google.com/drive/1nO0MMx0arjd01Pf04JKlX4C9bbIVsWzY?usp=sharing>

Github:

<https://github.com/Corv123/Machine-Learning-Proj>



Business Understanding

Problem Objective

- Predict median waiting time (hours) for admission from Emergency Department to hospital ward
- Enable hospitals to forecast patient flow and optimise resource allocation

Decision Context

Stakeholder

- Hospital operations: Staff scheduling, bed management
- Patients: Expectation setting, hospital selection
- MOH: Resource allocation across public hospitals

Success Criteria

- Primary: MAE < 3 hours (beat naive mean baseline)
- Secondary: Model explains meaningful variance ($R^2 > 0.3$)
- Operational: Predictions within realistic range



Data Understanding

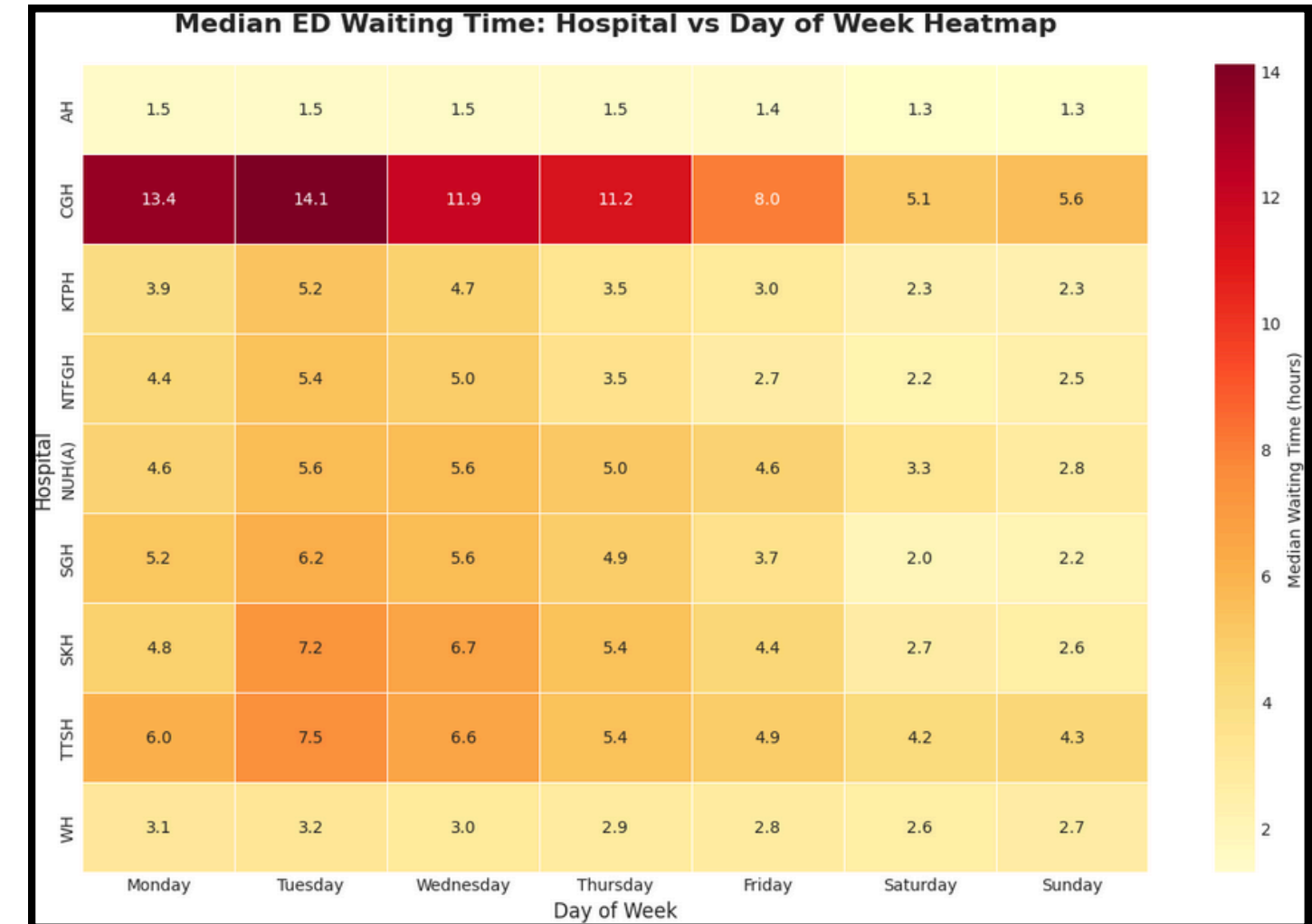
Target Variable

- Waiting time: Median hours from ED arrival to ward admission

Final Dataset

- 9,415 rows x 7 columns (after cleaning)
- 9 Hospitals: Singapore Hospital, Alexandra Hospital, Tan Tock Seng Hospital etc.

Dataset	Granularity	Period
Waiting time for admission to ward	Daily x Hospital	Jan2023 - Jan2026
ED Attendance	Daily x Hospital	Jan2023 - Jan2026
Bed Occupancy Rate	Daily x Hospital	Jan2023 - Jan2026
Public Holidays	Date List	2023 - 2026



Date	Day	Hospital	Waiting Time	Attendance	BOR	Is_Holiday
1/1/2023	Sunday	AH	1.6	64	80.5	1
1/1/2023	Sunday	CGH	7.7	351	89.6	1
1/1/2023	Sunday	KTPH	2.9	286	96.4	1
1/1/2023	Sunday	NTFGH	7.9	252	91.1	1
1/1/2023	Sunday	NUH(A)	2.1	257	78.3	1
1/1/2023	Sunday	SGH	1.3	309	79.3	1
1/1/2023	Sunday	SKH	3	333	86	1
1/1/2023	Sunday	TTSH	3.8	336	91.2	1
2/1/2023	Monday	AH	2	61	84.8	0
2/1/2023	Monday	CGH	12.7	386	90.6	0
2/1/2023	Monday	KTPH	12.2	326	97.9	0
2/1/2023	Monday	NTFGH	4.2	314	92	0

Data Understanding

Waiting Times (50th Percentile) for Admission from ED									
Date	AH	CGH	KTPH	NTFGH	NUH(A)	SGH	SKH	TTSH	WH
Sun, 01/01/23	1.6	7.7	2.9	7.9	2.1	1.3	3.0	3.8	
Mon, 02/01/23	2.0	12.7	12.2	4.2	2.5	2.5	2.7	4.4	
Tue, 03/01/23	2.1	16.0	12.9	22.8	4.0	9.3	5.0	4.9	
Wed, 04/01/23	2.5	16.3	21.4	14.7	8.2	11.6	10.3	6.9	
Thu, 05/01/23	2.4	21.5	20.7	18.0	5.2	11.7	7.9	5.0	
Fri, 06/01/23	2.8	18.3	20.0	8.2	6.4	6.8	4.8	4.9	
Sat, 07/01/23	1.3	11.5	6.8	2.4	3.7	1.7	7.7	5.1	
Sun, 08/01/23	1.2	5.4	2.8	2.6	6.1	1.7	5.0	4.8	
Mon, 09/01/23	3.3	11.1	17.9	6.7	5.4	7.6	13.9	5.4	
Tue, 10/01/23	9.0	17.8	6.2	9.7	5.3	6.6	15.0	4.7	
Wed, 11/01/23	2.2	18.3	6.7	12.0	4.8	6.1	16.3	4.8	
Thu, 12/01/23	2.4	14.0	11.4	9.5	5.4	3.1	5.6	4.6	
Fri, 13/01/23	1.4	7.9	3.5	5.5	3.8	2.0	8.3	5.2	
Sat, 14/01/23	1.2	3.6	4.8	2.5	2.3	0.8	5.2	4.1	

Issue: Missing values

Finding: Woodlands Hospital data unavailable before June 2024

Action: Dropped

Assumptions

- 1.Waiting time is accurately recorded at 50th percentile (median)
- 2.Hospital operational patterns remained relatively stable
- 3.Historical patterns are predictive of future waiting times

Fri. 18/04/25	0.4	4.6	2.8	2.7	3.3	1.5	
Mon. 22/12/25	1.7	21.5	3.3	2.9	4.6	6.4	5.2
Mon, 27/03/23	3.1	12.9	40				

Issue: Outliers

Finding: Waiting time ranges 0.4 - 40.0hrs+

Action: Retained (Reflects real variance → Admitted into Emergency department does not equal to admitted in a ward)

Risks

- 1.Temporal leakage: Used only same-day features; no future data
- 2.Distribution shift: Model may degrade if policies/ capacities change

Data Preparation

Cleaning steps:

- 1. Parsed date formats across three source files
- 2. Aligned to common date range (Jan 2023 - Jan 2026)
- 3. Merged on Date using inner join
- 4. Dropped 539 rows (5.4%) with missing values (WH pre-June 2024)

```
Common starting date (latest of the three starts)
common_start = max(wt["Date"].min(), emd["Date"].min(), bor["Date"].min())
print("Common starting date:", common_start.date())
```

```
Inner merge on Date so the final dataset only contains dates present in all three
combined = wt_f.merge(emd_f, on="Date", how="inner").merge(bor_f, on="Date", how="inner")
combined = combined.sort_values("Date").reset_index(drop=True)
```

```
Drop rows with missing values (WH before June 2024)
final_df = final_df.dropna(subset=['Waiting Time', 'Attendance', 'BOR'])
```

Encoding

Variable	Type	Encoding
Hospital	Categorical (9 levels)	One-hot (drop_first=True)
Day	Categorical (7 levels)	One-hot (drop_first=True)
Is_Holiday	Binary	Used as-is
Attendance, BOR	Numeric	Used-as-is

Train-Test Split

- 80% train / 20% test
- Random state = 42 for reproducibility

Baseline Modelling

Dummy Baseline

- DummyRegressor (mean strategy)
- Establishes minimum performance threshold

Features

- Hospital, Day, Attendance, BOR, is_Holiday

Classical Models (Default parameters)

- Linear Regression
 - Assumes linear relationships
 - Fast and interpretable
- Decision Tree Regressor
 - Captures non-linear patterns
 - Models feature interactions

Baseline Results

Performance Comparison

Model	MAE	RMSE	R ²
Dummy(mean)	3.358706	4.741237	-0.000571
Linear Regression	2.439310	3.666226	0.401722
Decision Tree	2.825199	5.002954	-0.114083

Key Findings

- Linear Regression outperforms other models
- 27.4% MAE improvement over dummy baseline
- MAE of 2.44 hours = typical prediction error
- R2 of 0.40 = explains 40% of variance

Critical Insight

- Decision Tree has negative R2 (0.11)
- Performs worse than dummy baseline
- Indicates overfitting with default params
- Strong justification for Stage 2 tuning

Initial Evaluation

Sanity Checks

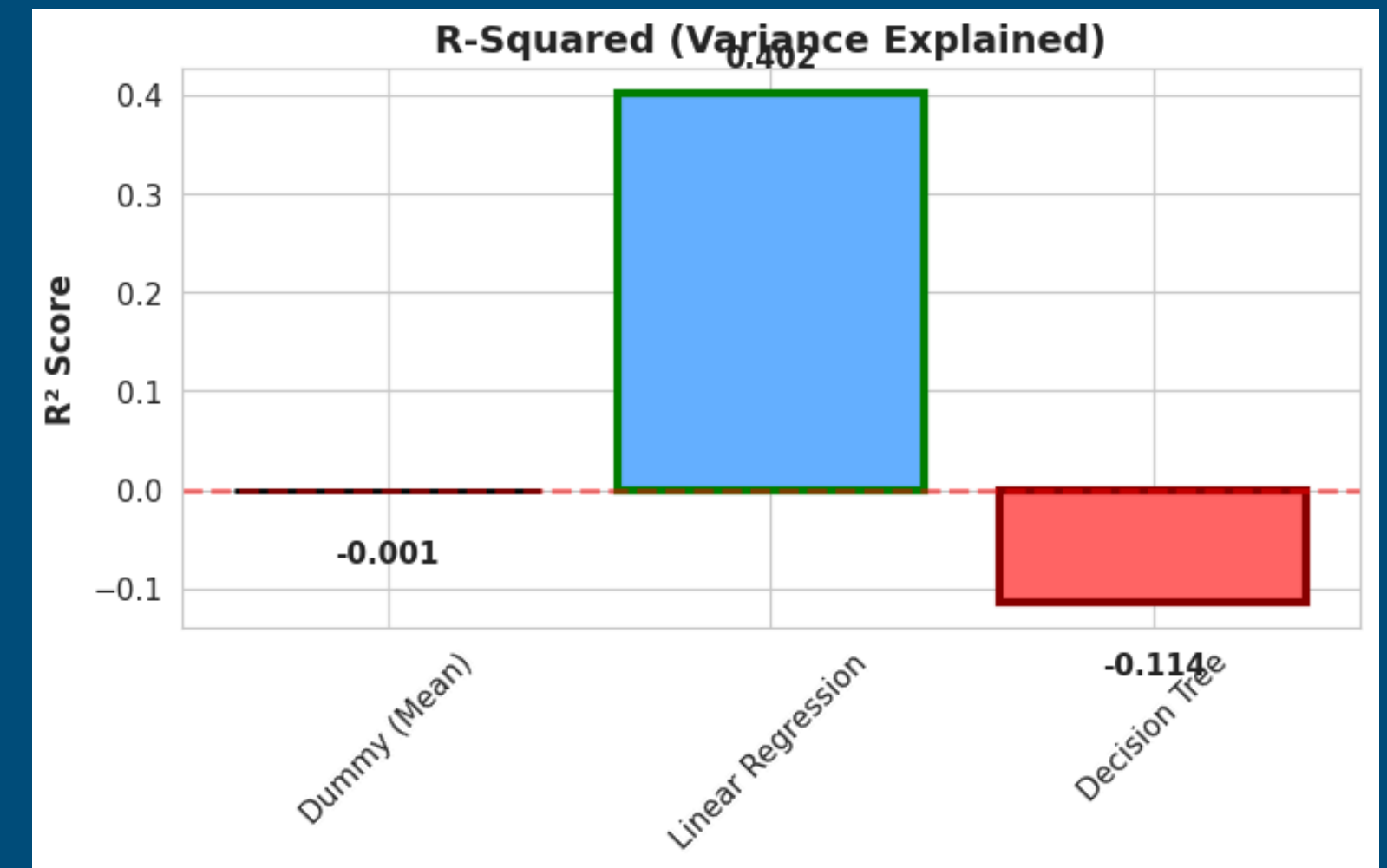
- 1) Linear regression had 27.4% improvement over baseline.
- 2) Predictions within realistic range

Why Did Decision Tree Fail?

- No max_depth limit = overfitting
- Memorized training data, can't generalize
- Needs hyperparameter tuning in Stage 2

Key Insights

- Linear relationships work well for this data
- 40% of variance explained ($R^2 = 0.40$)
- Average prediction error: 2.4 hours
- 60% variance unexplained → room for improvement



```
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SANITY CHECK 3: Feature Importance (Linear Regression)
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Top 10 features by absolute coefficient:

      Feature      Coefficient
Hospital_TTSH      -3.363767
Hospital_NTFGH      -2.843702
Hospital_SGH        -2.594089
Hospital_SKH        -2.049186
Hospital_CGH         1.942564
Hospital_NUH(A)     -1.823283
Hospital_KTPH       -1.649992
Day_Tuesday         1.365899
Hospital_WH         -1.317513
Day_Sunday          -1.312802
```


Next Steps

Modelling Improvements

Action	Rationale
Tune Decision Tree (max_depth, min_samples, leaf)	Fix overfitting observed in baseline (negative R ²)
Try Random Forest	Ensemble of trees for better generalisation
Ridge/Lasso Regression	Handle multicollinearity from one-hot encoding

Feature Engineering

Feature	Description
Lagged waiting time	Previous 1,3,7-day averages
Delta Occupancy	Change in BOR vs 48-hr average
Rolling ED attendance	3-day moving average of ED visits

External Data Sources

Data	Rationale
Infectious disease rates (flu/COVID)	More actionable for hospital managers
Air quality (PSI/PM2.5)	Haze periods drive respiratory admissions in Singapore

Alternative Problem Framing

Feature	Description
Multi-class classification	Predict Low/Medium/High wait hours

References

References

- MOH : <https://www.moh.gov.sg/others/resources-and-statistics/>
- Singapore Public Holidays: data.gov.sg

Tools Used

- Google Colab

Links

- Colab Notebook: <https://colab.research.google.com/drive/1nO0MMx0arjd01Pf04JKIX4C9bbIVsWzY?usp=sharing>
- Github: <https://github.com/Corv123/Machine-Learning-Proj>

Individual Contributions

Name/ID	Contribution
Ang Xuan Yu Pamela 2401870	Created EDA visualisations including hospital-day heatmap; identified waiting time extremes and patterns
GOH JING WEN 2403444	Implemented baseline models (Dummy, Linear Regression, Decision Tree); conducted train-test split; generated comparisons table
Hayden Chua Shao En 2402425	Developed sanity check framework; performed feature importance analysis; evaluated model predictions for realistic ranges
Jeanie Cherie Chua Yue-Ning 2403263	Completed Business Understanding and Data understanding sections; compiled slide deck
Corvan Chua 2402844	Developed data cleaning pipeline; parsed and merged data sources; handled missing value